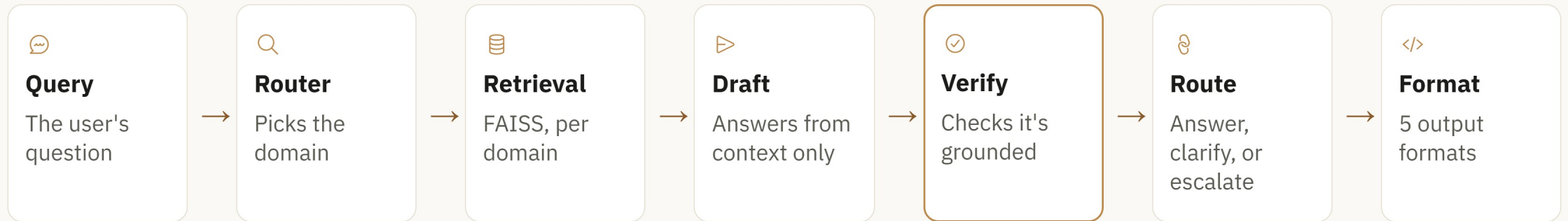


How a query becomes a trusted answer



Multi-turn memory carries the last few exchanges with every query, so a follow-up like "what about parental leave?" resolves against the previous question instead of starting over.

Built to run on nothing but the free tier

LLM	Gemini gemini-3.5-flash-lite, via google-genai
Embeddings	sentence-transformers, all-MiniLM-L6-v2 — local, no API call
Vector search	FAISS, one index per domain
Frontend	Streamlit, custom theme
Hosting	Streamlit Community Cloud, free tier

The original plan split work across two model tiers — a stronger model for drafting, a lighter one for routing and verification. That fell apart the first time the eval harness actually ran.

20/day
the stronger model's free quota
— burned through mid-run

\$0
total spend — every call now
runs on Flash-Lite

Classify, draft, and verify were consolidated onto Flash-Lite, plus a client-side limiter pacing requests apart. The pipeline itself didn't change — moving to a stronger model on a paid plan is a one-line edit.

A second, independent check before anything ships

ANSWER

The draft is grounded in the source chunks — ships with citations attached.

CLARIFY

The missing detail is about the user's own situation, not something the sources could list — so it asks, instead of guessing.

ESCALATE

No grounded answer is possible — including a correct refusal — so it hands off to a human team.

What this caught during testing

BUG CAUGHT

A correct refusal, marked as a confident answer

The draft accurately said it didn't have enough information — but verification still rated it grounded and high-confidence. Fixed with an explicit rule: a refusal is never an "answer."

BUG CAUGHT

Asked instead of just listing the answer

The knowledge base already listed every approval limit by role, but the draft asked which role the user was instead of answering. Fixed by teaching it to enumerate a fully-covered variable.

TRACK 3 — UNIFIED ENTERPRISE AI AGENT

Why this matters beyond the grade

4

domains covered — HR,
Finance, Support, Legal

48

knowledge-base documents,
authored by hand

75%

eval pass rate across 20 hand-
written test cases

0

hallucinated facts observed,
across every run

- ♥ Doesn't answer HR, Legal, or Finance questions with false confidence — the main risk of putting an LLM in front of employees and customers.
- 🗨 Escalates to a human only when it's actually unsure, not on every question.
- 🔑 Runs today on free infrastructure; moving to a stronger, paid model is a config change, not a rewrite.
- 🌐 Deployed and live — a reviewer can try it from a browser, no local setup required.

**Zero cost to build. Zero hallucinations observed.
One line away from a stronger model.**